

Peter Rodziňák

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Links: [Github](#)

About me

Master's degree student (BSc. successfully completed) focusing on digital design and embedded systems. I design hardware in Verilog and SystemVerilog, primarily develop in C++, and use Python for rapid prototyping. Seeking an opportunity (internship / part-time) to connect these domains on real-world projects in chip design and hardware development.

Top Projects

Advanced cryptography on FPGA (Basys 3)

Source: <https://github.com/PeterRodzinak/Portfolio/tree/main/Projects/Hardware%20design>

Tech stack: Verilog / SystemVerilog, C++, Python, FPGA (Xilinx Artix-7 / Basys 3), Vivado

- Design and complete implementation of the RC6 cryptographic algorithm directly at the hardware level.
- Successful timing closure resolution, logic element optimization, and chip area minimization while maintaining high throughput.
- Mathematical proof of concept and test vector generation initially implemented as a rapid Python prototype, subsequently translated into RTL description.

Application system on LilyGo TWatch-S3

Source: <https://github.com/PeterRodzinak/Portfolio/tree/main/Projects/Embedded%20system%20application>

Tech stack: C++, CMake, Git

- Hobby multi-application system for opensource smart-watch, managing WiFi connection, using third-party web services, time synchronization using NTP, etc.
- Implementation emphasizing battery optimization, application scalability, memory management, and object-oriented design.
- Utilization of modern C++ features and integration of automated project builds using a PlatformIO script.

Technical skills

- **Hardware Design:** Verilog, SystemVerilog, RTL coding, basics of functional verification, FPGA workflow (Basys 3, Vivado).

- **Software Development:** C++ (advanced concepts, low-level development), Python (basic scripting, prototyping).
- **Tools:** Git (version control), Linux/Unix environment, CMake

Education

Master's degree: Applied informatics, spec. Embedded Systems.

CTU FIT, Prague | Expected graduation: 2028

Bachelor's degree: Informatics, spec. Computer Engineering

CTU FIT, Prague | Graduated: 2025

Bachelor thesis topic: Robotic arm with tactile sensors

Work experience

- **Part-time job: Instructor/Animator** | Hemisféra I'avá o.z., Bratislava [2017 - 2022]

Teaching children and youth computer game development in programming and game design, supervising participants in a day camp.